

Optimization speed control of linear induction motors using CSA algorithm and nonlinear disturbance observer

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This paper investigates a high-performance robust control for linear induction motor (LIM) drives based on the combination of the Crow Search Algorithm (CSA) and a Nonlinear Disturbance Observer (NDO) within the Field-Oriented Control (FOC) framework. The dynamic behavior of LIMs is strongly affected by longitudinal end-effects, parameter variations, and external disturbances, which degrade the performance of conventional PI-based controllers. To overcome these drawbacks, the CSA metaheuristic is employed to optimally tune the speed PI controller gains by minimizing a transient performance index, ensuring a balanced trade-off between fast response and stability. In addition, a nonlinear disturbance observer is designed to estimate lumped disturbances arising from load variations, model uncertainties, and end-effect nonlinearities, allowing their compensation within the control loop. The proposed CSA–NDO–FOC control structure is developed using the LIM dynamic model expressed in the synchronous (d)–(q) reference frame while explicitly considering the end-effect phenomenon. The effectiveness of the proposed approach is validated through detailed simulations in the MATLAB/Simulink environment and compared with a conventional PI–FOC scheme. The obtained results demonstrate significant performance improvements under sudden load disturbances and parameter variations, including a reduction in overshoot from 9.33% to 4.33%, a decrease in settling time by approximately 37.5%, and an improvement in steady-state accuracy of nearly 76%. These results confirm that the combined CSA-based optimal tuning and NDO-based disturbance compensation provide a robust and effective solution for improving the dynamic performance and disturbance rejection capability of LIM drives in high-precision linear motion applications.

Key words: linear induction motor, field-oriented control, crow search algorithm, non-linear disturbance observer, longitudinal end effects, robust control

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1. Introduction

Modern control systems are inherently subject to external disturbances, unmodeled dynamics, and parameter variation, which collectively degrade performance, particularly in electric drives. These factors necessitate advanced disturbance estimation and compensation techniques, since traditional robust control approaches although effective at attenuating disturbances typically require conservative tuning and do not explicitly estimate disturbances, thereby restricting the simultaneous achievement of fast dynamic response and strong robustness [1]. In electric drives for linear motion applications, disturbances arising from load variations, friction, and structural effects play a dominant role. Linear Induction Motors (LIMs) introduce further complexities due to the longitudinal end-effect caused by the finite primary length, leading to significant nonlinearities, spatial variations in electromagnetic fields, thrust reduction at higher speeds, increased steady-state errors, and reduced efficiency. Field-Oriented Control (FOC) provides an effective decoupled framework for flux and thrust regulation; however, its performance remains highly sensitive to parameter variation and external perturbations, highlighting the critical need for robust disturbance rejection methods that preserve structural simplicity and implementation feasibility [2, 3]. Recent research has advanced nonlinear disturbance observer (NDO)-based strategies to address these challenges in induction motors. Notable developments include cascade nonlinear observer-based speed-sensorless adaptive twisting sliding mode control for LIMs to ensure precise speed tracking and robust compensation against end effects and uncertainties, novel NDO schemes for annihilating uncertain load torque and distortion effects in sensorless induction motors within direct field-oriented frameworks while improving transient response [4], adaptive current compensation with NDO for single-sided LIMs to mitigate dynamic eddy-current effects and load vibrations, high-order sliding-mode observer-based twisting controllers in LIMs, effectively handling dynamic end effects with reduced chattering [5, 6], nonlinear-disturbance-observer-based predictive control for trajectory tracking in planar/linear motor systems [7], time-varying state observer-based twisting control for LIMs considering dynamic end effects and unknown load torque [8], improved predictive current control using extended state observers for disturbance estimation in LIM drives [9], velocity disturbance rejection control via rolling prediction matrix for high-speed LIMs [10], finite-set model predictive thrust control to alleviate end-effect-induced thrust reduction and normal force variations [11]. Parallel to these model-based disturbance observers, Neural Network Observers (NNOs) have emerged as data-driven alternatives for disturbance estimation in electric drives. Approaches based on RBF, RNN, and LSTM networks have demonstrated accurate disturbance estimation and adaptability

to nonlinear dynamics without requiring explicit mathematical models [12, 13]. However, these methods generally require extensive training data, high computational resources, and do not provide the stability guarantees and interpretability offered by analytically designed disturbance observers.

Alongside advances in disturbance estimation and compensation techniques, metaheuristic optimization algorithms have gained significant attention in controller design due to their ability to efficiently tune parameters in complex nonlinear systems [14, 15]. In recent research studies, swarm intelligence algorithms have been widely employed for controller optimization in electric drives because of their strong global search capability and independence from gradient information. For instance, the Particle Swarm Optimization (PSO) algorithm, inspired by bird flocking behavior, has been extensively applied to PID tuning in induction motor drives, demonstrating superior convergence compared to gradient-based methods [16]. The Artificial Bee Colony (ABC) algorithm, modeling honeybee foraging strategies, has shown robust performance in multi-objective optimization of PI controllers for permanent magnet synchronous motor drives under parameter variation [17]. The Grey Wolf Optimizer (GWO), mimicking the leadership hierarchy of grey wolves, has been employed for fractional-order PI controller tuning in linear motor applications, achieving improved transient response [18]. The Whale Optimization Algorithm (WOA) and Firefly Algorithm (FA) have also been successfully applied to optimal torque control and efficiency optimization in induction and synchronous motors [19]. These swarm methods share common advantages: they do not require gradient information, handle non-convex and discontinuous search spaces effectively, and exhibit strong global exploration capabilities. Among these approaches, the Crow Search Algorithm (CSA), introduced by Askarzadeh [20], has attracted considerable interest due to its simplicity and efficient balance between exploration and exploitation. Inspired by the intelligent foraging behavior of crows, including food hiding, location memory, tracking, and awareness mechanisms, CSA employs a memory-based search strategy and adaptive flight length to effectively explore the search space while avoiding local optima. These characteristics make it particularly suitable for the multimodal optimization of LIM controller parameters.

CSA has been successfully applied in various engineering fields, including power system optimization, optimal reactive power dispatch, and controller parameter tuning. Recent developments have extended its application to electric drive systems, such as torque ripple minimization in switched reluctance motor drives, parameter identification, and advanced control optimization in motor and energy systems [21]. These promising results motivate the adoption of CSA in the present work for the optimal tuning of the proposed LIM control scheme. Motivated by these progresses in NDO-based disturbance estimation and meta-

heuristic optimization, the main contribution of this work is the integration of a Nonlinear Disturbance Observer (NDO) with CSA-based controller tuning to improve disturbance rejection, robustness, and dynamic performance in Linear Induction Motor (LIM) drives. This approach overcomes limitations of conventional methods, such as conservative robust tuning and vulnerability to unmodeled LIM dynamics, to deliver superior disturbance rejection, enhanced speed tracking accuracy, greater robustness across varying conditions, and optimized overall performance in a unified, computationally efficient structure. The work advances high-precision linear motion control, offering significant potential for industrial applications demanding reliable operation under inherent uncertainties.

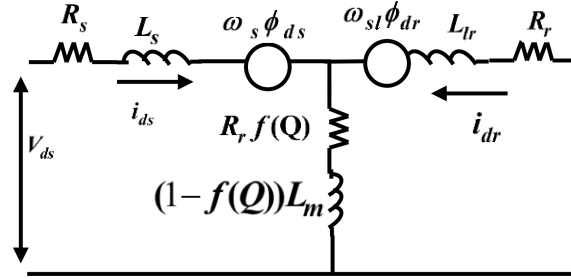
The remainder of this paper is organized as follows: Section 2 develops the mathematical model of the linear induction motor (LIM) in the synchronous reference frame and details the complete field-oriented control (FOC) architecture, including flux estimation and current regulation. Section 3 outlines the design methodology for the nonlinear disturbance observer (NDO), while Section 4 details the optimization process using the Crow Search Algorithm (CSA). Section 5 presents the simulation results and performance comparisons. Finally, Section 6 provides concluding remarks and discusses prospects for future hardware implementation.

2. Dynamic modeling and control strategy of the LIM

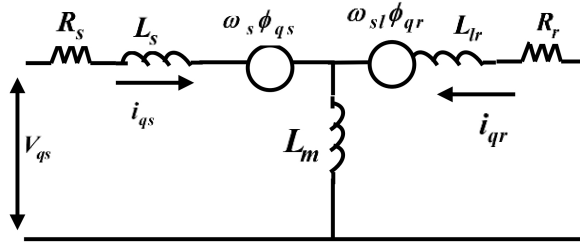
This section presents the mathematical modeling of the linear induction motor (LIM) and the control strategy based on field-oriented control (FOC). The modeling is developed in the synchronous rotating reference frame to facilitate control design. Then, the proposed FOC scheme is structured in a cascaded configuration composed of three hierarchical control loops. The outer speed loop generates the reference quadrature current i_{qs}^* from the speed error. The flux control loop regulates the direct current i_{ds}^* to maintain the desired rotor flux level. These loops are designed with fast dynamics to guarantee precise tracking of their respective reference signals i_{ds}^* and i_{qs}^* , ensuring stability and high dynamic performance of the overall control system. Several studies [22–24] in the literature demonstrate that the dynamic model of the LIM explicitly incorporates the longitudinal end-effect phenomenon through the introduction of a correction factor $f(Q)$ in the synchronous $d-q$ reference frame. This factor modifies the electromagnetic parameters and reflects the influence of the finite length of the primary on the air-gap flux distribution. Accordingly, the modified equivalent circuit of the LIM depicted in Figure 1 is derived by integrating the parameter Q ,

which characterizes the end effects [25], as follows:

$$Q = \frac{D \cdot R_r}{(L_m + L_{lr}) \cdot v} . \quad (1)$$



(a) direct-axis



(b) quadrature-axis

Figure 1: LIM equivalent electrical circuit

When eddy current losses are taken into account, an additional resistance appears in the transverse branch, expressed as:

$$\hat{R}_r = R_r f(Q), \quad (2)$$

where: $f(Q)$ is a correction factor defined as:

$$f(Q) = \frac{1 - e^{-Q}}{Q} . \quad (3)$$

The magnetizing inductance affected by the end effects is given by:

$$\hat{L}_m = L_m [1 - f(Q)] . \quad (4)$$

The electrical angular speed is related to the linear velocity by:

$$v = \frac{\tau_p}{\pi} \omega_r . \quad (5)$$

The stator and rotor voltage equations in the synchronous reference frame are expressed as:

$$\begin{cases} V_{ds} = R_s i_{ds} + R_r f(Q) (i_{ds} + i_{dr}) + \frac{d\phi_{ds}}{dt} - \omega_s \phi_{qs} , \\ V_{qs} = R_s i_{qs} + \frac{d\phi_{qs}}{dt} + \omega_s \phi_{ds} , \\ 0 = R_r i_{dr} + R_r f(Q) (i_{ds} + i_{dr}) + \frac{d\phi_{dr}}{dt} - \omega_{sl} \phi_{qr} , \\ 0 = R_r i_{qr} + \frac{d\phi_{qr}}{dt} + \omega_{sl} \phi_{dr} . \end{cases} \quad (6)$$

The flux-current relationships are defined as:

$$\begin{cases} \phi_{ds} = L_{ls} i_{ds} + L_m (1 - f(Q)) (i_{ds} + i_{dr}) , \\ \phi_{qs} = L_{ls} i_{qs} + L_m (i_{ds} + i_{dr}) , \\ \phi_{dr} = L_{lr} i_{dr} + L_m (1 - f(Q)) (i_{ds} + i_{dr}) , \\ \phi_{qr} = L_{lr} i_{qr} + L_m (i_{ds} + i_{dr}) . \end{cases} \quad (7)$$

The electromagnetic thrust force generated by the LIM is expressed as:

$$F_e = \frac{3\pi}{2\tau_p} \frac{p}{2} (\phi_{ds} i_{qs} - \phi_{qs} i_{ds}) . \quad (8)$$

The mechanical dynamics of the mover are described by:

$$F_e = M\dot{v} + B\dot{v} + F_L . \quad (9)$$

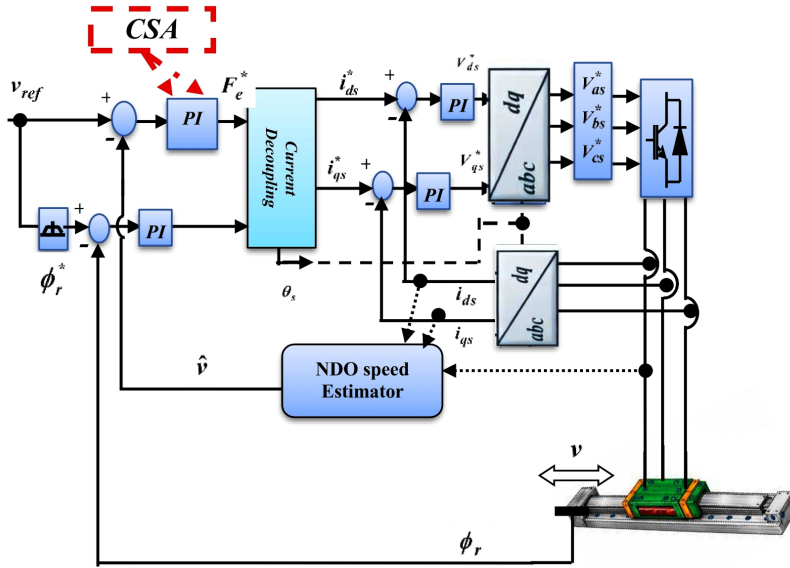
Here, M represents the mass of the moving parts, B is the damping coefficient and F_L is the load force.

To achieve decoupled control of rotor flux and electromagnetic thrust in the LIM, a field-oriented control (FOC) strategy is employed, as illustrated in Figure 2. In this strategy, the rotor flux vector is aligned with the d -axis of the synchronous rotating reference frame. This alignment results in the following relationships:

$$\begin{cases} \phi_{qr} = 0 , \\ \phi_{dr} = \phi_r = \text{constant} . \end{cases} \quad (10)$$

Under this condition, the electromagnetic thrust expression is simplified and can be written as:

$$F_e^* = K_f \cdot i_{qs}^* , \quad (11)$$



where K_f is a constant defined by:

$$K_f = \frac{3\pi}{2\tau_p} \frac{p}{2} \frac{L_m (1 - f(Q))}{L_r - L_m f(Q)} \phi_{dr}^* . \quad (12)$$

$$v_{sl} = \frac{\tau_p}{\pi} \cdot \frac{R_r}{L_r} \cdot \frac{\hat{L}_m \cdot i_{qs}^*}{\phi_{dr}^*}. \quad (13)$$

To eliminate cross-coupling between the d and q axes, feedforward compensation terms are introduced:

$$\begin{cases} E_d = -\frac{\hat{R}_r}{L_r}\phi_r + \omega_s\sigma' i_{ds}, \\ E_q = -\omega_s\sigma L_s i_{ds} - \omega_s\frac{\hat{L}_m}{L_r}\phi_r. \end{cases} \quad (14)$$

After compensation of the coupling terms, the stator voltage equations become:

$$\begin{cases} V_{ds}^* = \left(R_s + \hat{R}_r - \frac{\hat{R}_r \cdot \hat{L}_m}{L_r} \right) i_{ds} + \sigma L_s \frac{di_{ds}}{dt}, \\ V_{qs}^* = R_s i_{qs} + \sigma' \frac{di_{qs}}{dt}. \end{cases} \quad (15)$$

This decoupling results in two independent first-order subsystems, which significantly simplifies the control design and enhances the dynamic response of the overall system.

3. Nonlinear Disturbance Observer design

The Nonlinear Disturbance Observer (NDO) represents a sophisticated estimation method designed for the reconstruction of unknown disturbances acting on dynamic systems [26]. In contrast to conventional observers, which are strongly dependent on exact system models, the NDO conceptualizes all uncertainties, external perturbations, and unmodeled dynamics as a single lumped disturbance, which is then estimated through nonlinear error feedback.

3.1. Mechanical dynamics of LIM

The dynamic behavior of the LIM in the mechanical subsystem can be expressed as:

$$M \dot{v} = F_e - F_L - F_{\text{fric}} + d_{\text{end}} + d_{\text{param}}, \quad (16)$$

where F_e is the electromagnetic force, F_L is the load force, and F_{fric} represents friction effects. The terms d_{end} and d_{param} denote the disturbances caused by the end-effect and parameter uncertainties, respectively.

By normalizing the equation, the speed dynamics can be rewritten as:

$$\dot{v} = \frac{1}{M} F_e + d_m, \quad (17)$$

where d_m is the lumped disturbance defined as:

$$d_m = -\frac{F_L + F_{\text{fric}}}{M} + d_{\text{end}} + d_{\text{param}}. \quad (18)$$

3.2. Single Nonlinear Disturbance Observer design

To estimate and compensate for the lumped disturbance d_m , a reduced-order NDO is designed based solely on the speed dynamics (see Figure 3).

The observer is defined as:

$$\dot{\hat{v}} = \frac{1}{M} F_e + \hat{d}_m, \quad (19)$$

where $\hat{v}$ is the estimated speed and $\hat{d}_m$ is the estimated disturbance.

The disturbance estimation law is constructed using a nonlinear function of the estimation error:

$$\hat{d}_m = -k\phi(e_v), \quad (20)$$

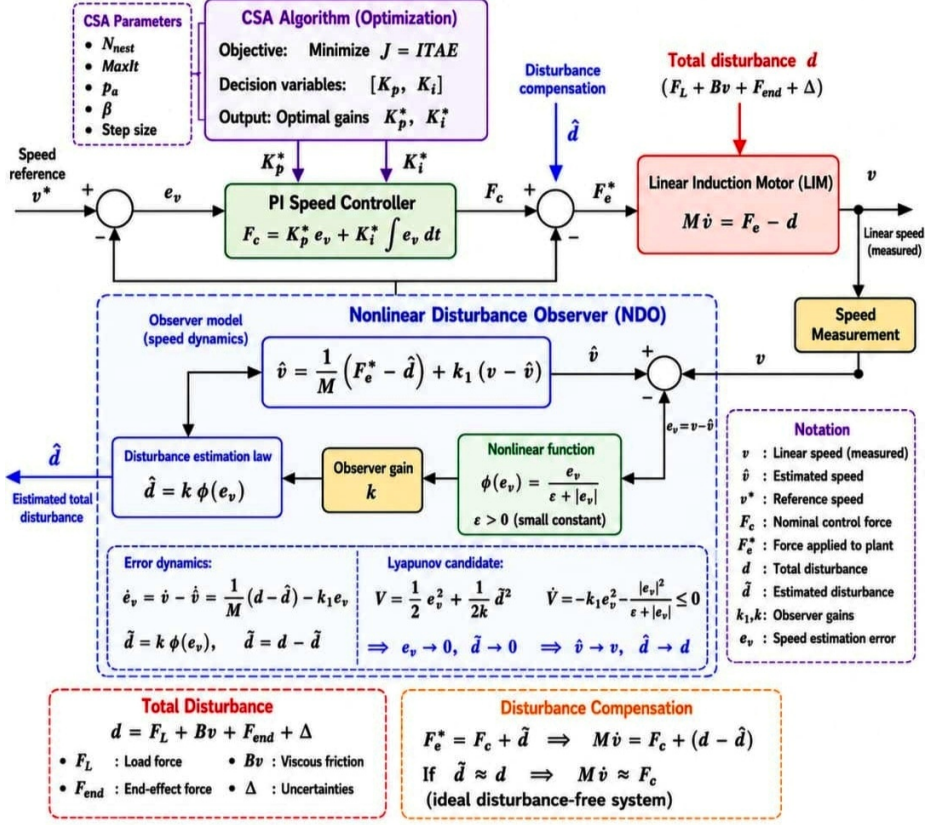


Figure 3: Block diagram of the Nonlinear Disturbance Observer (NDO) integrated within the CSA-optimized FOC control structure for LIM drives

where $k > 0$ is the observer gain and e_v is the speed estimation error given by:

$$e_v = v - \hat{v}. \quad (21)$$

To avoid chattering and ensure smooth estimation, the nonlinear function $\phi(\cdot)$ is selected as:

$$\phi(e_v) = \frac{e_v}{|e_v| + \varepsilon}, \quad (22)$$

where $\varepsilon > 0$ is a small constant.

Thus, the complete NDO formulation is given by:

$$\begin{cases} \dot{\hat{v}} = \frac{1}{M} F_e + \hat{d}_m, \\ \hat{d}_m = -k \frac{v - \hat{v}}{|v - \hat{v}| + \varepsilon}. \end{cases} \quad (23)$$

The observer gain k directly governs the convergence rate of the disturbance estimate via Eq. (20). A higher k reduces the estimation settling time but amplifies measurement noise. In this work, k is co-optimized with the PI controller gains using the CSA algorithm, ensuring an optimal balance between disturbance rejection bandwidth and noise attenuation.

3.3. Stability analysis

From the estimation error equation (21), the error dynamics can be derived as:

$$\dot{e}_v = d_m - \hat{d}_m. \quad (24)$$

Substituting the disturbance estimation law yields:

$$\dot{e}_v = d_m + k \frac{e_v}{|e_v| + \varepsilon}. \quad (25)$$

Consider the Lyapunov candidate function:

$$V = \frac{1}{2} e_v^2. \quad (26)$$

Its time derivative is:

$$\dot{V} = e_v \dot{e}_v = e_v d_m + k \frac{e_v^2}{|e_v| + \varepsilon}, \quad (27)$$

If the observer gain is selected such that:

$$k > |d_m|_{\max}. \quad (28)$$

Then $\dot{V} < 0$, which guarantees the convergence of the estimation error, i.e.,

$$e_v \rightarrow 0 \quad \text{and} \quad \hat{d}_m \rightarrow d_m. \quad (29)$$

3.4. Integration into FOC scheme

The estimated disturbance is incorporated into the speed control loop to enhance robustness. The reference electromagnetic force is modified as:

$$F_e^* = K_p (v^* - v) + K_i \int (v^* - v) dt - \hat{d}_m. \quad (30)$$

4. Crow Search Algorithm optimization

In recent years, the development of robust and optimal control strategies has become a major research focus, particularly for complex nonlinear systems such as Linear Induction Motors (LIMs), which are inherently affected by parameter

variation, external disturbances, and end-effect phenomena. To address these challenges, metaheuristic optimization techniques have emerged as powerful tools for controller tuning. In this context, the Crow Search Algorithm (CSA), proposed by Askarzadeh (2016) [20], has attracted considerable attention.

In the present study, CSA is employed to optimally tune the proportional and integral gains of the speed PI controller, and NDO gain k . Each crow represents a candidate solution in a three-dimensional search space:

$$x_i^t = [K_p, K_i, k]^T. \quad (31)$$

Each crow maintains a memory of its best position found so far: m_i^t .

The position of crow i is updated according to the following rule:

$$x_i^{t+1} = \begin{cases} x_i^t + r_i \cdot fl_i \cdot (m_i^t - x_i^t), \\ \text{a randomly generated position within the search bounds,} \end{cases} \quad (32)$$

$r_i, r \sim U(0, 1)$, fl_i , is the flight length, and $AP_j^t \in [0.1, 0.5]$ is the awareness probability of crow j .

After evaluation, the memory is updated if the new solution yields a better fitness value:

$$m_i^{t+1} = \begin{cases} x_i^{t+1}, & \text{if } J(x_i^{t+1}) < J(m_i^t), \\ [3pt] m_i^t, & \text{otherwise.} \end{cases} \quad (33)$$

The iterative process continues until the maximum number of iterations or a convergence criterion is met. The best memory position provides the optimal controller parameters (K_p^*, K_i^*, k^*). Figure 4 illustrates the complete flowchart of the CSA optimization process applied to the LIM speed controller tuning.

4.1. Objective function

The performance of each candidate solution is evaluated using the ITAE (Integral of Time-weighted Absolute Error) criterion, which is widely employed in dynamic system control to promote a fast response with minimal steady-state error, as defined by:

$$J = \int_0^{T_{\text{sim}}} t \cdot |v^*(t) - v(t)| dt. \quad (34)$$

The optimization problem is formulated as:

$$\min_{K_p, K_i, k} (J) \quad (35)$$

subject to the predefined bounds on K_p , K_i , and k .

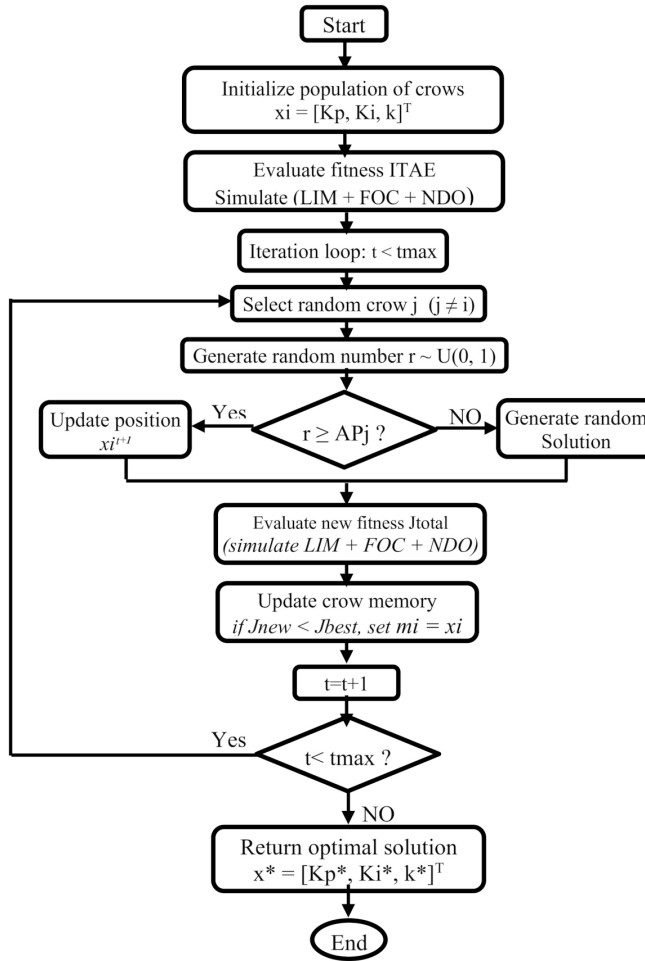


Figure 4: Flowchart of the CS algorithm for optimal PI speed controller tuning in LIM drives

From Figure 5, the fitness function (ITAE) decreases rapidly during the early iterations and converges to a minimum value of approximately 0.2555 after about 20 iterations. This result confirms the fast convergence and effectiveness of the CSA algorithm in identifying the optimal controller parameters, which are listed in Table 2 (see Appendix).

5. Results and discussions

This section presents a detailed evaluation and validation of the proposed CSA–NDO approach for Field-Oriented Control (FOC) of the Linear Induction Motor (LIM), with explicit consideration of end-effects. The effectiveness of the

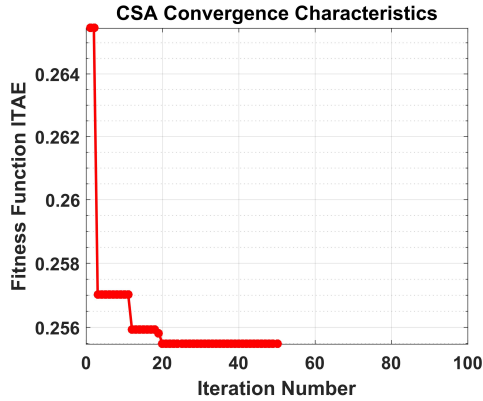


Figure 5: Evolution of the fitness function during CSA optimization

proposed control strategy is assessed through two principal operational scenarios: trajectory tracking accuracy and robustness against external disturbances. All simulations were conducted within the MATLAB/Simulink environment. The physical parameters of the LIM model are defined as follows: $R_s = 0.5 \, \Omega$, $R_r = 0.3 \, \Omega$, $L_s = 0.02 \, \text{H}$, $L_r = 0.015 \, \text{H}$, $L_m = 0.01 \, \text{H}$, and $\tau_p = 0.1 \, \text{m}$. To ensure optimal performance of the FOC scheme, the gains of the Proportional–Integral (PI) controllers were systematically tuned using the Cuckoo Search Algorithm (CSA). The corresponding CSA parameters used for the tuning process are summarized in Table 3 (see Appendix).

5.1. Test tracking performance

In this test, the simulations were running under dynamic operating conditions. These conditions included a step change in reference speed from 3 to 4 m/s at $t = 4 \, \text{s}$ and a sudden load force disturbance (FL) of 50 N applied at $t = 1.0 \, \text{s}$. The performance of three control strategies is compared: (1) conventional PI-FOC with fixed gains, (2) NDO-FOC with a nonlinear disturbance observer but without CSA optimization, and (3) the proposed CSA–NDO–FOC, which combines CSA-optimized PI gains with the NDO. As clearly demonstrated by the linear velocity response profiles illustrated in Figure 6, the actual velocity tracks the desired reference trajectory with high precision and minimal steady-state error for all three controllers. However, significant differences emerge in transient behavior. The conventional PI-FOC exhibits the largest overshoot (10.33%) and the longest settling time (1.0 s), accompanied by pronounced oscillatory behavior during speed transitions. The NDO-FOC improves upon this, reducing overshoot to 9.33% and settling time to 0.8 s, but still shows noticeable oscillations. In contrast, the CSA-optimized PI controller in the proposed CSA–NDO–FOC ensures a rapid

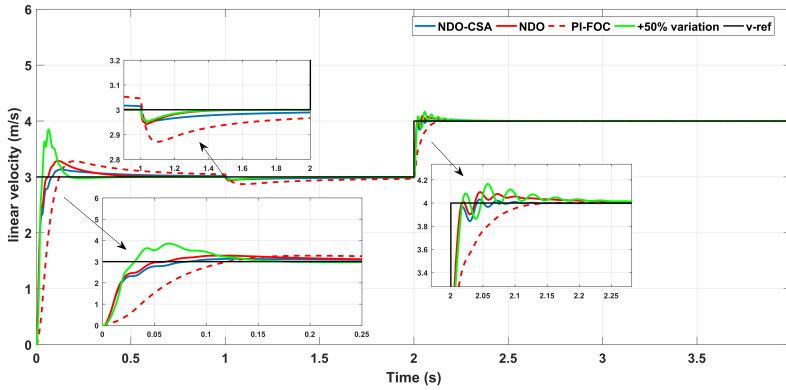


Figure 6: Comparative linear velocity response with +50% parameter variation

rise time (0.081 s) while maintaining a well-damped response, thereby effectively minimizing oscillatory behavior. The overshoot is reduced to only 4.33%, representing a 53.6% improvement over NDO-FOC and a 58.1% improvement over conventional PI-FOC. This confirms that the CSA-based optimization process successfully balances responsiveness and stability. The impact of the nonlinear disturbance observer (NDO) is clearly evident during disturbance events. When a sudden load of 50 N is applied at $t = 1.0$ s, the proposed CSA–NDO–FOC maintains accurate tracking with minimal deviation (1.6% speed drop). The system quickly returns to the reference value with a short recovery time. The NDO-FOC shows a moderate speed drop of 3.3%, while the conventional PI-FOC exhibits the most significant speed drop (4.3%) and the slowest recovery process, reflecting its sensitivity to nonlinearities and lack of disturbance compensation. This confirms that the disturbance observer is capable of estimating and compensating lumped disturbances, including those arising from end-effect nonlinearities and external perturbations, and that CSA tuning further enhances this capability by providing optimal controller gains.

Figure 7 depicts the electromagnetic thrust force response, which represents the most challenging dynamic characteristic of the entire system. At startup and during the step change in reference speed, a significant force spike is produced across all three controllers, driven by the sudden increase in transient power demand. This behavior is physically consistent with the inertial and resistive forces that the linear motor must overcome during acceleration phases. The conventional PI-FOC produces the largest force spikes and exhibits the most sustained oscillations, while the CSA–NDO–FOC demonstrates the fastest damping and smallest force overshoots, confirming its superior dynamic control.

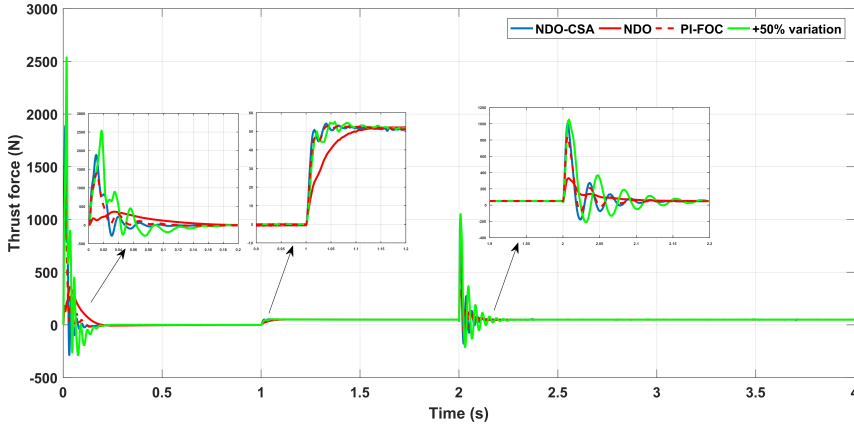


Figure 7: Electromagnetic thrust force response with +50% parameter variation

The direct (i_{ds}) and quadrature (i_{qs}) current responses shown in Figure 8 demonstrate significant improvements in current regulation through the CSA–NDO approach. Notably, the quadrature current (i_{qs}), which directly influences the thrust force, exhibits enhanced tracking accuracy and stability. The conventional PI-FOC shows the most sluggish i_{qs} response with higher ripple, while the CSA–NDO–FOC achieves the fastest convergence and lowest current ripple. Similarly, the direct current (i_{ds}), responsible for flux regulation, remains most stable in the proposed method, with minimal deviation from its reference value during transients.

The flux responses further validate the tracking performance of the proposed control strategy. As illustrated in Figure 9, the direct-axis rotor flux ϕ_{dr} is consistently regulated around its reference value (approximately 0.4–0.45 Wb) for all controllers, but the CSA–NDO–FOC achieves the tightest regulation with the smallest transient deviations. The conventional PI-FOC shows the largest flux dips during speed step changes, while the NDO-FOC improves upon this but still exhibits more deviation than the proposed method. In parallel, the quadrature-axis flux ϕ_{qr} remains tightly controlled near zero throughout the operation for all strategies, with the CSA–NDO–FOC demonstrating the best decoupling performance. These results confirm effective decoupling between flux and force components, which is a fundamental requirement of field-oriented control (FOC). Consequently, even under transient conditions, such as speed variations or load disturbances, any flux deviations remain minimal and are rapidly corrected in the proposed CSA–NDO–FOC, demonstrating its strong tracking performance.

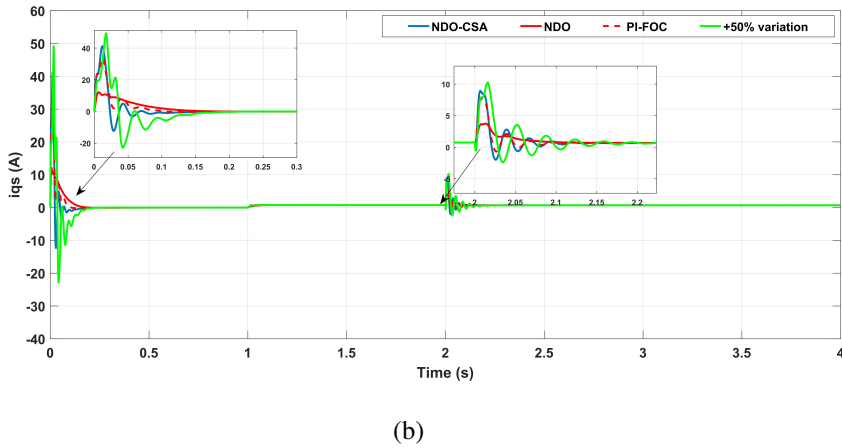
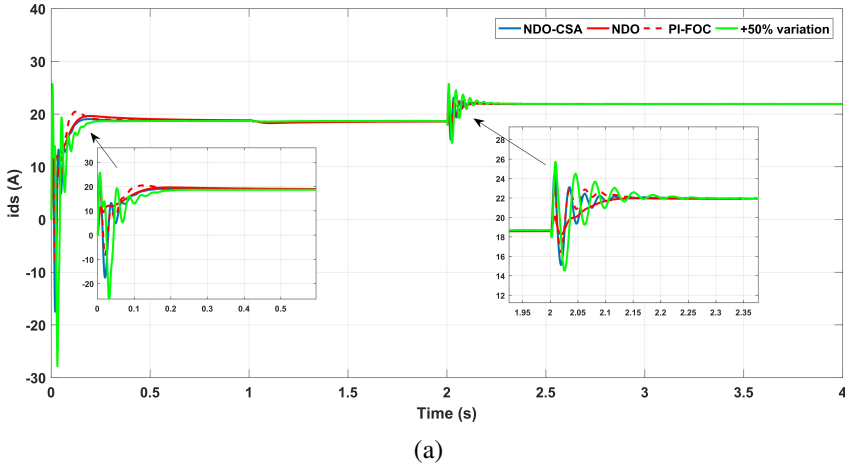
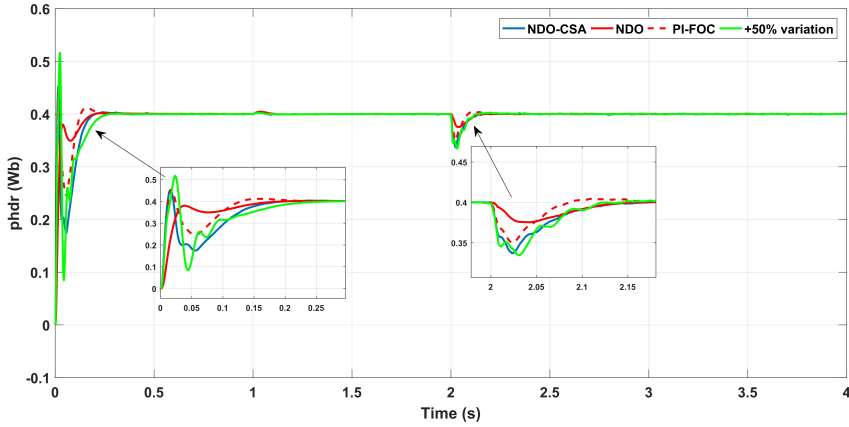


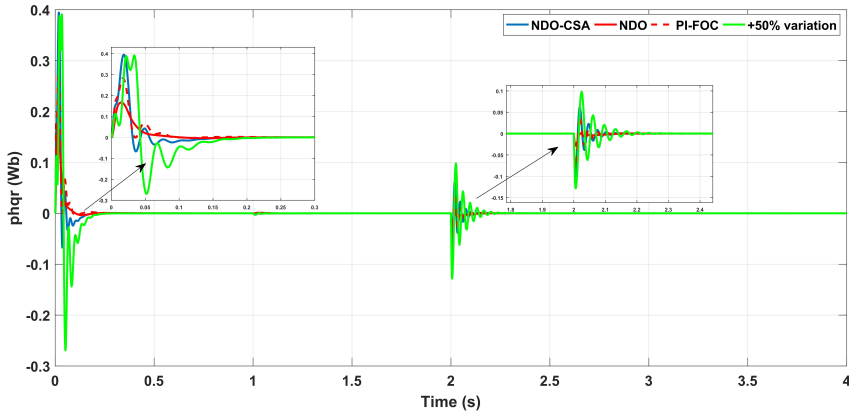
Figure 8: Stator current responses under +50% parameter variation: (a) direct-axis component (i_{ds}), (b) quadrature-axis component (i_{qs})

5.2. Test robustness

To further assess the robustness of the proposed CSA–NDO–FOC, a parameter variation test was carried out in which the stator resistance R_s and the motor mass M were simultaneously increased by +50% from their nominal values. Under these perturbed conditions, the performance of all three controllers was evaluated. The conventional PI-FOC showed severe degradation, with overshoot increasing dramatically and settling time extending significantly. The NDO-FOC demonstrated better robustness than PI-FOC, maintaining reasonable stability but with noticeable performance degradation. In contrast, the CSA–NDO–FOC



(a)



(b)

Figure 9: Comparative flux responses under +50% parameter variation: (a) direct-axis component (ϕ_{dr}), (b) quadrature-axis component (ϕ_{qr})

controller remained remarkably stable, with only small degradation in overshoot, settling time, and oscillatory behavior during speed transitions. This observation demonstrates the excellent robustness of the combined CSA–NDO approach to parameter variations and load disturbances, significantly outperforming both the conventional PI-FOC and the NDO-FOC without CSA optimization. To consolidate these observations quantitatively, Table 1 presents a direct comparison of key performance metrics among the three control strategies: conventional PI-FOC, NDO-FOC, and the proposed CSA–NDO–FOC, under both nominal and per-

turbed ($R_r+50\%$, $M+50\%$) conditions. This comparison offers a clear numerical perspective on the improvements discussed throughout this section.

Table 1: Quantitative performance comparison between PI-FOC, NDO-FOC, and proposed CSA-NDO-FOC

Performance metric	PI-FOC	NDO-FOC	CSA-NDO	$R_r+50\%$, $M+50\%$
Rise time (s)	0.11	0.121	0.081	0.04
Settling time (s)	1	0.8	0.5	0.4
Overshoot (%)	10.33	9.33	4.33	28.3
Steady-state error	0.067	0.05	0.012	0.055
Speed deviation (load disturbance)	4.3%	3.3%	1.6%	5%

The quantitative comparison summarized in Table 1 confirms the superior performance of the proposed CSA-NDO-FOC scheme over both conventional PI-FOC and NDO-FOC. Under nominal conditions, the proposed method achieves the fastest rise time (0.081 s), representing a 33.1% improvement over NDO-FOC (0.121 s) and a 26.4% improvement over conventional PI-FOC (0.11 s). Similarly, the settling time decreases to 0.5 s, which is a 37.5% reduction compared to NDO-FOC (0.8 s) and a 50% reduction compared to PI-FOC (1.0 s), reflecting improved damping and quicker convergence to steady state. A significant improvement is also observed in overshoot, which is reduced from 10.33% (PI-FOC) and 9.33% (NDO-FOC) to 4.33% in the proposed CSA-NDO-FOC. This represents a reduction of approximately 58.1% compared to PI-FOC and 53.6% compared to NDO-FOC, confirming that the proposed method effectively limits excessive transient peaks. In steady-state conditions, the error is reduced from 0.067 (PI-FOC) and 0.05 (NDO-FOC) to 0.012, highlighting the higher accuracy of the CSA-NDO approach. The speed deviation under load disturbance is reduced from 4.3% (PI-FOC) and 3.3% (NDO-FOC) to 1.6% (approximately 62.8% and 51.5% improvement, respectively), demonstrating enhanced disturbance rejection capability. Even under severe parameter variations ($R_r+50\%$, $M+50\%$), the CSA-NDO-FOC maintains acceptable performance, whereas the conventional PI-FOC would fail to maintain stability. Overall, these results confirm that the CSA-based tuning provides a well-balanced trade-off between fast response and system stability, while maintaining robustness under parameter variations.

6. Conclusions

This study proposes a robust and optimized control framework for Linear Induction Motor (LIM) drives, achieved through the integration of a Nonlinear Disturbance Observer (NDO) and Crow Search Algorithm (CSA)-based tuning within a Field-Oriented Control (FOC) scheme. The proposed structure addresses the inherent weaknesses of conventional PI-FOC, particularly in the presence of end-effects, parameter uncertainties, and external disturbances. The CSA was used to determine optimal controller gains, while the reduced-order NDO enabled real-time estimation and compensation of aggregated disturbances. The proposed strategy was evaluated under demanding operating conditions, including load force disturbances, parameter variations up to +50%, and end-effect nonlinearities, and was compared against both conventional PI-FOC and NDO-FOC without CSA optimization. Comparative analysis demonstrates that the proposed CSA-NDO-FOC controller significantly outperforms both the classical PI-FOC and the NDO-FOC strategies in terms of stability and tracking accuracy under these conditions. This superior performance is characterized by a 58.1% reduction in overshoot compared to PI-FOC (53.6% compared to NDO-FOC), while rise time, settling time, and steady-state error were improved by 26.4%, 50%, and 82.1%, respectively, compared to conventional PI-FOC. Furthermore, the speed response remained stable with minimal deviation (1.6%) during sudden load disturbances, highlighting the effectiveness of its disturbance rejection mechanism. The results obtained through MATLAB/Simulink confirm the superior performance of the CSA-NDO-FOC approach in both tracking and robustness tests. Its precise control, stability, and adaptability make it a preferred choice for linear induction motor applications, especially in dynamic and uncertain environments. Overall, the presented approach offers a practical and effective solution for high-performance LIM drives operating under uncertain and disturbed environments.

Appendix

Table 2: Comparison of conventional and CSA-optimized controller parameters

Parameters	Conventional controller	CSA-optimized controller
K_p	6.5	7.8184
K_i	49.15	55
K (NDO gain)	7	0.78

Table 3: CSA algorithm parameters and search space configuration

Parameter	Symbol	Value	Description
Number of crows	n_nest	40	Population size of candidate solutions
Maximum iterations	$MaxIt$	50	Total number of optimization iterations
Discovery rate	p_a	0.25	Probability of abandoning a nest
Lévy flight index	β	1.5	Controls step size distribution in Lévy flight
Step size scaling	–	0.01	Scaling factor applied to Lévy flight displacement
$[K_p, K_i, K]$		[0.1, 10 0.01] [20 100 10]	Lower bounds Upper bounds

Nomenclature

R_s, R_r	– Stator and rotor resistance (Ω)
L_s, L_r	– Stator and rotor inductance (H)
L_m	– Magnetizing inductance (H)
V_{ds}, V_{qs}	– Stator voltage components in the synchronous $d-q$ frame (V)
$i_{ds}, i_{qs}, i_{dr}, i_{qr}$	– Stator and rotor current components in the synchronous $d-q$ frame (A)
$\phi_{ds}, \phi_{qs}, \phi_{dr}, \phi_{qr}$	– Stator and rotor flux linkages (Wb)
ω_{sl}	– Synchronous electrical angular speed (rad/s)
τ_p	– Pole pitch (m)
p	– Number of pole pairs
Q	– Factor associated with the length of rotor
F_e, F_L	– Electromagnetic (thrust) and load force
D	– Length of the linor (m)
σ, σ'	– Leakage coefficients with and without end-effect

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